

Inflation Money

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An Introduction

I've finally came around to checking out Polymarket - a betting site on every possible subject. I wanted to try and learn about time-series analysis and trying a bet on inflation seemed like an easy entry¹

The Bet

I wanted to study and try out a bet with a few guiding principles:

1. Data is available and simple
2. The bet could be studied using simple time-series methods

I've stumbled around and found the following simple finance question:

How high will inflation get in 2026?

With the following odds:

\$945,213 Vol. Dec 31, 2026		Polymarket	
Above 4% \$228,871 Vol. 𐄂	97% ▲ 63%	Buy Yes 97.1¢	Buy No 3.0¢
Above 4.5% \$8,642 Vol. 𐄂	61% ▲ 11%	Buy Yes 61¢	Buy No 40¢
Above 5% \$137,773 Vol. 𐄂	32% ▲ 12%	Buy Yes 32¢	Buy No 69¢
Above 6% \$38,218 Vol.	17% ▼ 17%	Buy Yes 18¢	Buy No 84¢
Above 10% \$37,856 Vol.	6%	Buy Yes 7.9¢	Buy No 95.2¢
Above 8% \$37,109 Vol.	7%	Buy Yes 8.4¢	Buy No 93.6¢
Hide resolved ^			
Above 3% \$280,045 Vol.		Yes ✓	
Above 3.5% \$176,699 Vol.		Yes ✓	

¹in hindsight it's probably one of the toughest things to try and predict. The US government tries and fails at it constantly. Clearly, I'll manage to outsmart 'em.

We can see that inflation already crossed the 3% and 3.5% thresholds - they appear at the bottom as “resolved”, i.e whoever bet on these won. It’s important to understand the context of the bet - if at ANY month during the year inflation passes some threshold, it will be resolved in favor of whoever placed a bet on that threshold. In this problem’s context, a given month’s threshold is calculated using the *Consumer Price Index*. We need to get a little formal so we can start thinking about the problem:

Denote π_m the inflation ratio (denoted by IR from now on) at the end of month $m \in (1, \dots, 12)$. Denote $t \in (4, 4.5, 5, 6, 8, 10) := T$ the critical thresholds suggested in the bet. Our event of interest is defined as $E_t := \{\max_{m \in (1, \dots, 12)} \pi_m\} > t$, for all thresholds t . We would like to model the probability $P(E_t)$ for all $t \in T$

The general process will be composed of 2 stages:

1. Collect, organize and analyze initial data
2. Construct (learn and test) a model designed to predict the probability $P(E_t)$

Step 1 - The Data

After understanding what we would like to predict, we can start building some basic models. First, we need to gather some data. I’ve downloaded data about past CPI information². Each row is a single year, each column is a month. As always, we need to clean up the data before we can start using it effectively. The following code imports the data, and cleans it up so we can start processing it:

```
data <- read_excel("hist_cpi-u.xlsx", skip = 3)
data <- data[-1, ] # remove first row
data <- data[, -1] # remove first col
data <- data %>%
  slice(1:(n()-3)) %>% # remove last 3 lines
  mutate(across(everything(), as.numeric)) # make sure all columns are numeric
```

here are the first couple of rows to get a feel for what we have:

```
## # A tibble: 6 x 13
##   Year Jan. Feb. Mar. Apr. May Jun. Jul. Aug. Sep. Oct. Nov. Dec.
##   <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 1913  9.8  9.8  9.8  9.8  9.7  9.8  9.9  9.9  10   10   10.1  10
## 2 1914  10   9.9  9.9  9.8  9.9  9.9  10   10.2  10.2  10.1  10.2  10.1
## 3 1915  10.1  10   9.9  10   10.1  10.1  10.1  10.1  10.1  10.2  10.3  10.3
## 4 1916  10.4  10.4  10.5  10.6  10.7  10.8  10.8  10.9  11.1  11.3  11.5  11.6
## 5 1917  11.7  12   12   12.6  12.8  13   12.8  13   13.3  13.5  13.5  13.7
## 6 1918  14   14.1  14   14.2  14.5  14.7  15.1  15.4  15.7  16   16.3  16.5
```

Step 2 - The Models

Brief Introduction

Now let’s get our hands dirty with time-series analysis. In general, a time-series is just a sequence of data points indexed in chronological order. We can order our data in many different ways. One natural way to do so, is to go linearly in time. We can take $x_0 := PCI_{\text{January, 1913}}$, $x_1 := PCI_{\text{February, 1913}}$ and so on.

²<https://www.bls.gov/cpi/tables/supplemental-files/>

As we've seen earlier, what we're interested in is the inflation ratio - not the CPI of each month directly. From a short search in Google, we find the following relationship between IR and CPI at time t :

$$\pi_t := \frac{CPI_t - CPI_{t-12}}{CPI_{t-12}} \cdot 100$$

We should note that this is not a very comfortable variable, and a more suitable one could be $x_t := \log(CPI_t) - \log(CPI_{t-12}) = \log(\frac{CPI_t}{CPI_{t-12}})$. First, notice the direct connection between π_t and x_t . if we denote $r_t := \frac{CPI_t}{CPI_{t-12}}$, we have $\pi_t = (r_t - 1) \cdot 100$ and $x_t = \log(r_t)$. We can use Taylor series approximation $\log(x + 1) \approx x$ for small 'x's, and thus $\log(r_t) = \log(1 + (r_t - 1)) \approx r_t - 1$. Therefore: $\log(r_t) \approx \frac{\pi_t}{100} \rightarrow \pi_t \approx 100 \cdot \log(r_t)$

We can therefore model x_t and easily find the connection back to π_t . log-scales are preferred for a couple of reasons - they are easy to manipulate algebraically, and represent proportional changes well. From now on, the variable we would like to predict is the log difference - defined simply as $x_t := \log(CPI_t) - \log(CPI_{t-12})$.

Model 1 - Auto-Regressive

We start off with a simple Auto Regressive model of order p , defined as:

$$X_t = \sum_{i=1}^p \phi_i X_{t-i} + \epsilon_t$$

The model has the following assumptions:

1. $\epsilon_t \sim N(0, \sigma_\epsilon^2)$
2. ϕ_i are constants, $i \in (1, \dots, p)$

There are a couple of things worth checking before making predictions:

1. whether x_t is stationary process
2. the optimal order for a model

Stationary

In non-technical terms, a stationary process is a process whose statistical properties don't change over time. This is an important property we wish our process will have, as it gives us the ability to learn patterns from the past, and assume these patterns will continue in the future.

We start out by fitting an optimal ordered AR model:

```
# data edit
monthly_cpi <- pivot_longer( #
  data,
  cols = Jan.:Dec.,
  names_to = "Month",
  values_to = "CPI"
)

x_t <- diff(log(monthly_cpi$CPI), lag = 12) # x_t = log(CPI_t) - log(CPI_{t-12})
x_t <- na.omit(x_t)

# model fit
model <- ar(x_t)
```